



6954 - Spectroscopic confirmation of the most robust $z=15$ galaxy candidate

Cycle: 4, Proposal Category: GO

INVESTIGATORS

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OBSERVATIONS

<i>Folder</i>	<i>Observation</i>	<i>Label</i>	<i>Observing Template</i>	<i>Science Target</i>
Observation Folder				
	1		NIRSpec Fixed Slit Spectroscopy	(1) 3003658

ABSTRACT

We propose an efficient NIRSpec PRISM observation of the single most robust and luminous $z=15$ galaxy candidate selected from all available JWST NIRCам imaging - PANORAMIC- $z15$. Given that this is the only robust $z=15$ galaxy candidate detected in all of the current NIRCам imaging, spectroscopic verification of the redshift is vital for our understanding of the nature of star formation at cosmic dawn.

JWST Proposal 6954 (Created: Thursday, July 17, 2025, 9:00:11AM Eastern Standard Time) - Overview

This efficient 6-hour program will not only definitively confirm the redshift of this object via (at least) the spectroscopic detection of the Lyman break, but, crucially, it will also detect rest-frame UV emission lines. Specifically, our exposure time of 4 hours should enable the detection of CIII] 1908 given the typical minimum equivalent widths in similar sources at extreme redshifts but may also allow the detection of NIV 1488, CIV 1549, HeII 1640, NIII 1750 and potentially even Lyman-alpha. This will place crucial constraints on the nature of star formation and/or AGN activity only 270 Myr after the Big Bang. By spectroscopically confirming a galaxy at $z=15$, we will not only set a new redshift record but also reveal the nature of star formation earlier than ever measured before.

OBSERVING DESCRIPTION

We will observe the luminous $z=15$ candidate PANORAMIC- $z15$ using the NIRSPec/PRISM mode. We will use the fixed slit S200A1 to minimise the exposure time required due to the superior sensitivity compared to the MSA.

We will use the NRSIRS2 readout mode to keep the data volume within the required limit. WATA will be used for the target acquisition. For each exposure we will have 3 integrations with 22 groups. There will be 3 dithers leading to a total exposure time of 14574.30 s (4.1 hours). Once overheads are accounted for there is a total charged time of 5.9 hours.

Proposal 6954 - Targets - Spectroscopic confirmation of the most robust z=15 galaxy candidate

Fixed Targets	#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
	(1)	3003658	RA: 22 17 0.0855 (334.2503563d) Dec: +00 22 45.18 (.37922d) Equinox: J2000		
	Comments: Category=Galaxy Description=[High-redshift galaxies] Extended=YES				
	(2)	TA_star	RA: 22 16 58.8501 (334.2452087d) Dec: +00 22 57.42 (.38262d) Equinox: J2000		
	Comments: Category=Star Description=[G stars] Extended=NO				

Proposal 6954 - Observation 1 - Spectroscopic confirmation of the most robust z=15 galaxy candidate

Observation	Proposal 6954, Observation 1											Thu Jul 17 14:00:11 GMT 2025
	Diagnostic Status: Warning											
	Observing Template: NIRSpec Fixed Slit Spectroscopy											
Diagnostics	(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous		
	(1)	3003658	RA: 22 17 0.0855 (334.2503563d) Dec: +00 22 45.18 (.37922d) Equinox: J2000									
	Comments: Category=Galaxy Description=[High-redshift galaxies] Extended=YES											
Acquisition	#	Target	TA Method	Subarray	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID	
	1	2 TA_star	WATA	FULL	F110W	NRSRAPID	3	1	1	42.947	255813	
Template	HFF Readout Mode				Slit				Subarray			
	false				S200A1				FULL			
Dithers	#	Primary Dither Positions							Sub-Pixel Pattern			
	1	3							NONE			
Spectral Elements	#	Grating/Filter	Slit	Readout Pattern	Groups/Int	Integrations/Exp	#	Autocal	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
	1	PRISM/CLEAR	S200A1	NRSIRS2	22	3	1	NONE	3	9	14574.301	